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Quantum Machine Learning Based on K-Means to Optimize Cross-Border Intelligent Payment Supervision

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Abstract

With the rapid development of cross-border payment business, the existing regulatory mechanism faces challenges in dealing with complex payment data and transaction patterns, and traditional algorithms are difficult to efficiently deal with multi-dimensional transaction risks. By introducing quantum machine learning technology based on K-Means clustering, a new method for optimizing intelligent regulation of cross-border payments is proposed. This study combines the high parallelism of quantum computing with K-Means algorithm to analyze the risk characteristics in cross-border payment data, and verifies the effectiveness and reliability of the model in different scenarios through experiments. The results show that the quantum machine learning model can significantly improve the accuracy and efficiency of payment risk identification, and provide a more refined tool for supervision. However, the application of quantum computing still faces the dual challenges of technical realization and cost, and the deep integration of quantum computing and financial algorithms should be further explored in the future to improve the practicability and security of payment supervision.

CCS Concepts

• **Computer mothodologies** → Machine learning; Machine learning approaches; Kernel methods.

Keywords

Quantum Machine Learning, Algorithm, Banking, Payment Supervision, Cross-Border

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1 INTRODUCTION

In recent years, the intersection of quantum computing and machine learning has become a research hotspot, and the financial industry, as the core of the global economy, its security, efficiency and regulation are extremely important to social development. At present, cross-border bank payment systems are facing increasingly complex data processing needs, and traditional algorithms are gradually showing limitations in terms of processing speed, data scale and system complexity. Quantum computing provides a new path to solving these problems through parallel processing capabilities, along with efficient algorithms from quantum machine learning.

Although some scholars have proposed the application potential of quantum algorithms in the financial field, there is still a research gap in the specific application scenarios of bank payment systems. This is because there are still uncertainties in the development of the theoretical basis of quantum machine learning, and the effective integration of existing algorithmic frameworks with bank payment processes is not yet clear. The limitations of quantum computing hardware make its practical application face challenges, and the experimental verification is difficult. In this regard, how to overcome technical obstacles, improve the practicality of quantum machine learning algorithms, and optimize the application effect in bank payment system supervision has become a very important issue. Based on the K-Means theory, this study uses a combination of mathematical analysis and model construction to explore the potential application path of quantum machine learning in bank payment. [1]

By analyzing the application status of quantum machine learning in the field of bank payment, the paper fills the gap between the existing theory and practice, and provides effective decisions for bank supervision. Explore how quantum algorithms can play a role in improving the efficiency of payment regulatory systems, enhancing data security, and optimizing risk management. The research may provide a new technical path for the banking industry, not only help to improve the intelligence level of the payment system, but also help to promote the wide application of quantum computing in the financial field, but also provide theoretical support and technical reserves for the future combination of quantum computing and international financial law, and provide innovative solutions for industry practice.

2 PROBABILISTIC DENSITY ESTIMATION

Many quantum machine learning methods are based on sample sets of known class labels. However, in many practical applications, due to the lack of knowledge to form a pattern class process, or because of the difficulties in practical work, often can only use the sample set without class labels to work. This is commonly referred to as unsupervised learning. Generally speaking, unsupervised learning methods can be divided into two categories, namely direct method

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based on probability density function estimation and indirect clustering method based on similarity measurement between samples. The commonality of the two methods is that after dividing the sample set into several subsets, it can be used to solve the classification problem directly, or it can be used as the training sample set for classifier design.

For each subset, the probability density function is unimodal, and each subset is equivalent to a class. In order to avoid the difficulty of estimating the probability density function, the set can be divided into several subsets according to the similarity between samples under certain conditions, and the partition result should be the maximum of some criterion function representing the clustering quality. When distance is used to represent the similarity between two samples, the result is that the feature space is divided into several regions, each of which corresponds to a category. Some commonly used distance measures can be used as such similarity measures. The reason why distance is often used to represent the similarity between samples is that empirically, the feature vectors of samples of the same class should be close to each other, while the distance between the feature vectors of samples of different classes is much larger. Although this method seems to be different from the clustering method based on mixed probability density function estimation, the estimation by dry$p(y)$is also carried out on the basis of the distance between samples, and the two feature vectors that are very close to each other often belong to the same unimodal subset. Therefore, conceptually, it is still a clustering method for data sets, namely iterative dynamic clustering algorithm and non-generation hierarchical clustering algorithm. [2]

Density Estimation, is to estimate the probability density function of a random variable based on some observed samples. Dense Degree estimation is widely used in data modeling and machine learning. The density estimation methods can be divided into two types: parametric density estimation and non-parametric density estimation. Parametric density estimation is based on the assumption of randomness. The variables follow a certain distribution, and then the parameters of the distribution are estimated by training samples.

Let $D = \{x^{(n)}\}_{i=1}^{N}$ is N training samples independently extracted from an unknown distribution, assuming that the samples obey a probability distribution function $p(x|\theta)$, the log-likelihood function is

$$logp\left(D|\theta\right) = \sum_{n=1}^{N} logp\left(x^{(n)}|\theta\right). \tag{1}$$

We need to estimate a parameter θML such that

$$\theta^{ML} = arg_{\theta} max \sum_{n=1}^{N} logp\left(x^{(n)}|\theta\right). \tag{2}$$

The parameter estimation problem is then transformed into an optimization problem.

2.1 Normal Distribution

Hypothetical sample$x \in Rd$follows a normal distribution

$$\begin{array}{l} N\left(x|\mu,\Sigma\right) = \frac{1}{2(2\pi)^{d/2}|\Sigma|^{1/2}} \\ exp\left[-\frac{1}{2}\left(x-\mu\right)T\Sigma-1\left(x-\mu\right)\right] \end{array}. \tag{3}$$

where$\mu$and$\Sigma$are the mean and variance of the normal distribution, respectively. The log-likelihood function is

$$\begin{array}{l} logp\left(D|\mu,\Sigma\right) = -\frac{N}{2}\ log\left[(2\pi)^{2}\left|\Sigma\right|\right) \\ -\frac{1}{2}\Sigma_{n=1}^{N}(x-\mu)^{T}\Sigma^{-1}\left(x-\mu\right) \end{array} \tag{4}$$

The partial derivatives of the above equations with respect toμ,Σ, respectively, and make them equal to 0.[3] Can get,

$$\mu^{ML} = \frac{1}{N}\Sigma_{n=1}^{N}(x)^{(n)} \tag{5}$$

$$\Sigma^{ML} = \frac{1}{N}\Sigma_{n=1}^{N}\left(x-\mu\right)\left(x-\mu\right)^{T} \tag{6}$$

2.2 Multinomial Distribution

Assuming that the samples obey the multinomial distribution of K states, let the one hot vector $x \in [0,1]$ Kdenote thek^{th}State, i.e. $x_k = 1$, the rest $x_{i,i\neq k} = 0$. The probability density function for sample$x$is

$$p(x|\mu) = \pi_{K=1}^{K}\mu_k^x k \tag{7}$$

where$\mu k$is the probability of thek^{th}state, and satisfies

$$\Sigma_{k=1}^{K}\mu k = 1 \tag{8}$$

Dataset $D = \{x^{(n)}\}_{n=1}^{N}$, The log-likelihood function is

$$logp\left(D|\mu\right) = \Sigma_{n=1}^{N}\ \Sigma_{k=1}^{K}x_k^{(n)}log\left(\mu_k\right) \tag{9}$$

Parameter estimation of multinomial distribution is a constrained optimization problem. The Lagrange multiplier$\lambda$is introduced to turn the original problem to the unconstrained optimization problem.

$$max_{\mu,\lambda}\ \Sigma_{n=1}^{N}\ \Sigma_{k=1}^{K}x_K^{(n)}log\left(\mu_k\right) + \lambda\left(\Sigma_{k=1}^{K}\mu_k - 1\right) \tag{10}$$

Separate the partial derivative of the above equation with respect toμ_k,λ, and set it equal to 0. We may get,

$$\mu_k^{ML} = \frac{m_k}{N}, \quad 1 \leq k \leq K \tag{11}$$

Including$m_k = \Sigma_{n=1}^{N}x_k^{(n)}$is the number of samples whose value is thek^{th}state in the data set.

In practical applications, parameter density estimation generally has the following problems:

(1) Model selection problem: how to select the density function of data distribution. Distribution of actual data to be very complex, not a simple normal distribution or multinomial distribution.

(2) Unobservable variable problem: that is, the sample we use to train only contains some observable variables. There are also some very critical variables that cannot be observed, which makes it difficult to accurately estimate the truth of the data.

(3) Dimensional disaster problem: Parameter estimation of high-dimensional data is very difficult. As the dimensions increase, estimate

The number of samples required to calculate the parameters increases exponentially. Overfitting occurs when the sample is insufficient.[4]

The K-means algorithm is illustrated using the Old Faithful data set in Figure 1. For the purposes of this example, we have made a linear re-scaling of the data, known as standardizing, such that each of the variables has zero mean and unit standard deviation. For this

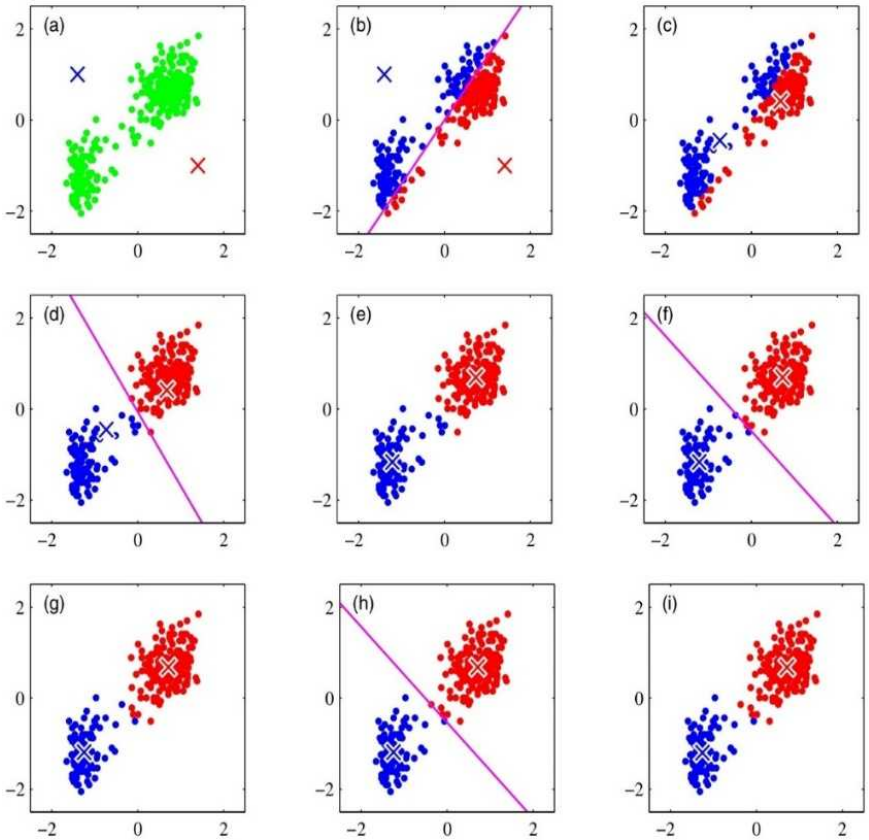

Figure 1: K-Means Clustering

example, we have chosen$k = 2$, and so in this case, the assignment of each data point to the nearest cluster center is equivalent to a classification of the data points according to which side they lie of the perpendicular bisector of the two cluster centers. A plot of cost function J given by for the Old Faithful example is shown in Figure 2.

Illustration of the K-means algorithm using the re-scaled Old Faithful data set. (a) Green points denote the data set in a two-dimensional Euclidean space. The initial choices for centers$\mu_1$and$\mu_2$are shown by the red and blue crosses, respectively. (b) In the initial E step, each data point is assigned either to the red cluster or to the blue cluster, according to which cluster center is nearer. This is equivalent to classifying the points according to which side of the perpendicular bisector of the two cluster centers, shown by the magenta line, they lie on. (c) In the subsequent M step, each cluster center is re-computed to be the mean of the points. Assigned to the corresponding cluster. $(d) - (i)$show successive$E$and Msteps through to final convergence of the algorithm.

Figure 3. Plot of the cost function$J$given by after each$E$step (blue points) and$M$step (red points) of the K-Means algorithm for the example shown in Figure 1. The algorithm has converged after the third M step, and the finalEMcycle produces no changes in either the assignments or the prototype vectors.[5]

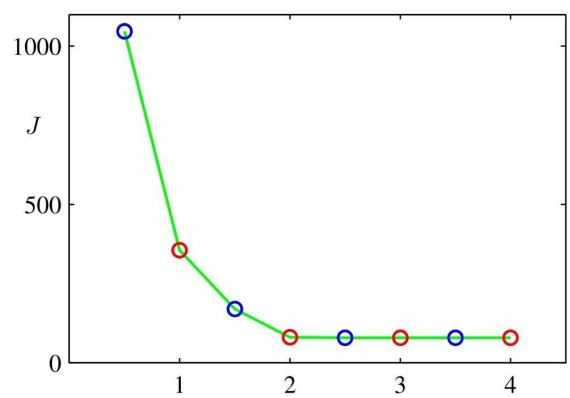

Figure 2: Old Faithful Example

3 NONPARAMETRIC DENSITY ESTIMATION

Nonparametric density estimation does not assume that the data obeys a distribution approximated by dividing the sample space into different regions and estimating the probability of each region, based on the probability density function.

For one random vector$x$in high dimensional space, suppose it obeys an unknown distribution$p(x)$, then x falls, the probability of

the small region R into the space is

$$P = \int Rp\left(x\right) dx \tag{12}$$

Given$N$training samples$D = \{x^{(n)}\}_{N=1}^{N}$, the number of samples Kfalling into the region Rfollows the binomial distribution

$$PK = \binom{N}{K} P^{K}\ \left(1 - P\right)^{1-K} \tag{13}$$

where$\frac{K}{N}$is expected to be$E[\frac{K}{N}] = P$, the variance is var$(\frac{K}{N}) = \frac{P(1-P)}{N}$. When$N$ is not

When it is constant, we can approximately think that

$$P \approx \frac{K}{N}. \tag{14}$$

Assuming that the region$R$is small enough that probability density within it is the same, there is

$$P \approx p\left(x\right) V \tag{15}$$

where$V$is the volume of the region$R$. Combining the above two formulas, we obtain

$$p\left(x\right) \approx \frac{K}{NV} \tag{16}$$

According to Equation, an accurate estimate of$p(x)$requires that the sample number$N$be as large as possible, The domain volume$V$is as small as possible. However, in specific applications, the number of samples is generally limited, and the area is too small. This results in fewer samples falling into this region, which makes the estimated probability density less accurate. Therefore, real two methods are usually used for nonparametric density estimation: (1) Fixed area size$V$, statistics fall into different number of regions, including histogram method and kernel method. (2) Change the area size so that the number of samples to fall into each region is$K$, which is called the K-nearest neighbor method. [6]

3.1 Histogram Method

The Histogram Method[1] is a very intuitive way to estimate the density function of continuous variables. The number method can be expressed as a bar chart.

Taking a one-dimensional random variable as an example, its value range is first divided into$M$continuous, non-overlapping intervals (bin), the width of each interval is Δm. GivenΔNtraining samples $D = \{x^{(n)}\}_{n=1}^{N}$, we count these samples fall into the number Km of each interval and then normalize them to a density function.

$$p_m = \frac{K}{N\Delta_m}, \quad 1 \le m \le M \tag{17}$$

where the interval width Δm is usually set to the same value Δ. The key problem of histogram method is how to choose one. A suitable interval width Δ. If Δ is too small, then the number of samples falling into each interval will be smaller, and its estimated interval density also has great randomness. If delta is too large, its estimated density function becomes ten. It is very difficult to reflect the real data distribution. Figure 3 shows an example of histogram density estimation, which the middle blue line shows the true density function, and the red bar chart shows the density estimated by the histogram method. [7]

Histograms are often used to deal with low-dimensional variables and visualize the distribution of data very quickly. The disadvantage is that it is difficult to scale to high-dimensional variables. Suppose a random vector of d dimensions, if each dimension is divided$M$intervals, then the number of intervals in the entire space isM^d. Number of samples required for histogram method will exponentially increase as dimension d increases, resulting in a Curse of Dimensionality Problem.

3.2 Kernel Method

Kernel Density Estimation, also known as the Parzen window method, is a method of estimation Improved histogram method.

Suppose R is a "hypercube" centered on point x in a D-dimensional space, and define the kernel function

$$\Phi(\frac{z-x}{h}) = \{ \begin{smallmatrix} 1 & if|z_i - x_i| < \frac{h}{2}, \ 1 \le i \le d \\ 0 & else \end{smallmatrix} \tag{18}$$

To indicate whether a sample z falls into the hypercube, where h is the side length of the hypercube, also known as the nucleus the width of the function.

Given$N$training samples$D = \{x^{(n)}\}_{n=1}^{N}$, and the number of samples$K$falling into region$R$is

$$K = \Sigma_{n=1}^{N} \Phi\left(\frac{x^{(n)} - x}{h}\right) \tag{19}$$

Then the density of point$x$is estimated to be

$$P\left(x\right) = \frac{K}{Nh^d} = \frac{1}{Nh^d}\ \Sigma_{n=1}^{N} \Phi\left(\frac{x^{(n)} - x}{h}\right) \tag{20}$$

whereh^drepresents the volume of the region$R$.

In addition to the kernel function of the hypercube, we can also choose smoother kernel functions, such as height Kernel function,

$$\Phi\left(\frac{z-x}{h}\right) = \frac{1}{2\pi^{\frac{1}{2}} h}\ exp\left(-\frac{||z-x||^2}{2h^2}\right) \tag{21}$$

whereh^2can be viewed as the variance of a Gaussian kernel function. So the density of point x is estimated to be[8]

$$p\left(x\right) = \frac{1}{N}\Sigma_{n=1}^{N} \frac{1}{(2\pi)^{\frac{1}{2}} h}\ exp\left(-\frac{||z-x||^2}{2h^2}\right) \tag{22}$$

3.3 K-Nearest Neighbor Method

The kernel width in the kernel density estimation method is fixed, so the same width may be for the high-density region. The domain is too large and the area of low density is too small. A more flexible way is to set a variable width of the area, and has resulted in a sample fall into each area of fixed number of$K$. To estimate the density of point$x$, first find one. A sphere centered on$x$, such that the number of samples falling into the sphere is K, and then according to the formula, We can calculate the density of point$x$. Because the samples that fall into the sphere are also the$K$samples closest to$x$, so this method is called the K-Nearest Neighbor method.[2]

[1]Histogram is derived from the Greek histos (to erect) and gramma (to trace) by the British statistician Carl Pearson proposed it in 1895.

[2]The K-nearest neighbor method is not a strict one for density function estimation methods.

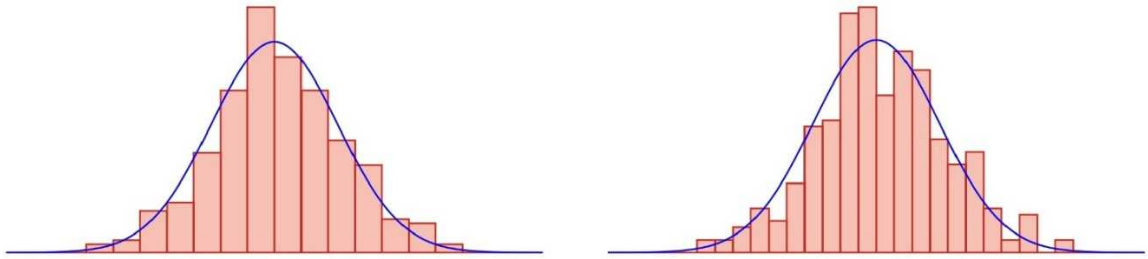

Figure 3: (a) Histogram Density Estimation (10 bins); (b) Histogram Density Estimation (30 bins)

In K-nearest neighbor method, the choice of$K$is also very important. If$K$is too small, the density cannot be estimated efficiently the degree function, and$K$too large will also make the local density inaccurate, and increase the computational overhead. [9]

K-nearest neighbor methods are also often used in classification problems and are called K-nearest neighbor classifiers. When$K = 1$is also called the most neighbor classifier. A property of the nearest neighbor classifier is that its classification error rate does not exceed when$N \to \infty$. More than twice the error rate of an optimal classifier. [10]

4 SIMULATION ANALYSIS

After discussing the various algorithms described above, we can see the data construction of the point set, the number of sample points in the analyzed point set, the distance measure and similarity measure used, the clustering criterion used, and the final number of clusters will affect the classification results. In addition, it is very important to determine the scale proportion of each component of the sample. In general, the indirect dynamic clustering algorithm is very efficient, so many people are happy to use it. Its main disadvantage is that the selected models often fail to reflect the probability structure of data, so the results obtained by these methods cannot reflect the real situation of data construction. We can only partially solve this difficulty by selecting various kernel functions and analyzing the clustering results obtained by using these kernel functions.

The separation algorithm of unimodal subset class needs to estimate the probability density function, so the calculation work is very heavy. In addition, because some parameters are selected in probability estimation, the estimation results are also greatly influenced by the parameters. Especially in the case of noise, the number of peaks of the probability density function with local maximum will change, which can not reflect the number of unimodal subsets in the data correctly. In the case of a small number of samples, it is impossible to estimate the probability density function, which makes this method completely meaningless. Hierarchical clustering algorithms may be particularly useful at this time.

One of the main reasons why there is more uncertainty in unsupervised pattern recognition problems than in supervised pattern recognition problems is that we do not have a sample set of known categories in unsupervised problems. We may not even know the number of categories, and the amount of information available is greatly reduced. In practical applications, in addition to what is described in this chapter, attention should also be given to finding effective use of expertise in the field of applications in order to fill information gaps. In the end, the actual meaning of the cluster is often explained and determined only by the relevant knowledge. [11]

5 CONCLUSION

This research discusses the optimal application of quantum machine learning based on K-Means clustering algorithm in cross-border smart payment supervision. With the increasing frequency of global cross-border payments, it is difficult for traditional regulatory models to efficiently process massive payment data, especially under the regulatory requirements of different countries, and the existing system often suffers from low processing efficiency and high false positive rate. Therefore, this study aims to improve the efficiency and accuracy of the regulatory system by combining the parallel processing capability of quantum computing with the pattern recognition advantages of K-Means clustering algorithm. The study first constructed a historical payment dataset to cluster transactions using classical K-Means, followed by a quantum computing simulation of the environment to test the performance of the quantum-optimized K-Means algorithm on large-scale data. The experimental results show that the quantum K-Means algorithm is superior to the traditional method in both data processing speed and accuracy of abnormal transaction recognition. Especially when processing millions of transaction data, quantum acceleration significantly improves the clustering efficiency and reduces the false positive rate. Although the method shows significant potential in theory and experiment, practical application still faces many challenges, such as quantum hardware is still immature, and regulatory standards in different countries are not uniform. Therefore, future research is needed to further optimize the algorithm and explore its application in different regulatory environments. The research in this paper reveals the great potential of quantum computing in financial regulation, points out the importance of algorithms advancing in tandem with policy, and emphasizes the application value of unsupervised learning in payment regulation. Future research directions will focus on the practical deployment of quantum algorithms, the integration of cross-border payment networks, and how to achieve global harmonized regulatory standards.

Acknowledgments

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